Astronomical Anomaly Detection with Spiking Neural Networks

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Abstract

In modern astronomical surveys, the generated time series data is extensive, and infrequent occurrences are indicative of important phenomena. Identifying anomalies is difficult because they are scarce, complex, and often lack sufficient annotated data for a comprehensive understanding. This project proposes a semi-supervised anomaly detection approach based on spiking neural networks (SNNs) for characterizing normal behavior in temporal signals and identifying anomalies indicative of anomalous events. Astronomical continuous light curves are represented as spike trains to enhance changes and enable the SNN to learn normal behavior. Synthetic time series signal data with anomalies are used for experiments to quantitatively measure performance. To prove the applicability of the proposed approach, it's tested on the available Kepler light curve data. Performance is evaluated based on detection rate, false alarm rate, and anomaly scoring consistency. The project's objective is to evaluate the effectiveness of SNNs for temporal anomaly detection in large-scale astronomical datasets.

1 Introduction

Time-domain astronomy is defined as the study of how astronomical objects change over time, encompassing any naturally occurring physical entities within the universe. From star clusters to galaxies, these systems consist of multiple astronomical bodies or substructures, such as asteroids or planets. The continuously evolving technology involved in astronomy has made the study of time-domain astronomy more feasible and important. Changes that were once too minuscule to observe can now be captured with high temporal resolution. This has permitted researchers to make a step forward in studying rare anomalies, as well as processes related to stellar evolution and planetary dynamics, from events that occur within a span of milliseconds to years [1]. Due to the extremely large temporal datasets in astronomy, this topic presents an opportunity to apply machine learning methods to efficiently identify anomalous patterns.

Anomaly detection plays a key role in astronomical surveys, which often produce datasets of unparalleled scale and greatly increase the probability of scientific discovery. Due to the sheer number and variance of sources, it is impractical to manually inspect or label each piece of data. As a result, traditional supervised learning approaches are infeasible. Methods capable of learning useful representations of astronomical anomalies have thus grown to become increasingly important [2].

Spiking neural networks (SNNs), in particular, provide a basis that mimics biological neural networks. SNNs have been built on artificial neural networks (ANNs) and deep neural networks (DNNs), representing information through discrete spike timings instead of continuous activations [3]. This structure creates an efficient modeling of sparse, time-varying signals that are found in astronomical observations such as light curves and RFI measurements [4]. SNNs have the capacity for efficient, real-time detection of anomalous patterns, which are overlooked by traditional machine learning methods.

2 Related Work

Recent surveys of SNN research demonstrate their application across multiple domains. In Sabbella et al.'s paper, the authors focus on SNNs for time-series analysis in ubiquitous computing applications, including human activity recognition, speech classification, physiological signal processing, and emotion recognition. The survey highlights several advantages of SNNs for temporal data, where their sparse, event-driven signalling allows for energy-efficient computation, and their spike-based representations naturally capture temporal dynamics more effectively than traditional machine learning models, such as Recurrent Neural Networks, Long Short-Term Memory networks (LSTMs), Convolutional Neural Networks (CNNs), and Transformers. SNNs have also shown strong, real-time processing capabilities, making them a promising approach for resource-constrained and temporal applications [5]. These properties suggest that SNNs are well-suited for analyzing astronomical time-series data, where anomalies may indicate scientifically rare, significant events.

Recent advances in machine learning have shown tremendous progress in the study of astronomical data. For example, Muthukrishna et al. developed RAPID, a real-time automated photometric time-series classification tool that identifies transient phenomena. RAPID employs deep recurrent neural networks with gated recurrent units for classifying astronomical time-series data. RAPID can be applied to ongoing surveys such as the Zwicky Transient Facility (ZTF) and the Large Synoptic Survey Telescope (LSST) [6]. Similarly, within the context of SNNs, Pritchard et al. demonstrate how automated systems can efficiently process dynamic spatio-temporal data through radio frequency interference (RFI) detection. Their study describes the potential of SNNs in time-series functions, achieving competitive performance while discovering meaningful patterns in astronomical datasets [7].

In terms of our project, there has been similar work that follows implementation steps comparable to our methodology. For example, Dennis Bäßler et al. propose an unsupervised anomaly detection method for multivariate time series using SNNs. Their work introduces a rank-order-based learning algorithm that utilizes the precise times of incoming spikes. In addition, their work describes an encoding technique based on multi-dimensional Gaussian receptive fields to handle temporal data. The authors similarly evaluate the effectiveness of their approach using synthetic datasets with various anomaly types, demonstrating strong, effective performance in detecting anomalies in streaming time series data [8].

3 Methodology

3.1 Pipeline Overview

Our methodology has been refined into a structured pipeline consisting of the following stages: data generation, preprocessing, spike encoding, SNN modeling, anomaly scoring, and evaluation. Since the proposal phase, we have evaluated multiple design choices and finalized an approach for each stage of the pipeline. We evaluated both synthetic and real-world astronomical datasets. Synthetic, astronomical-like time-series signals were used in the initial phase to simulate normal behavior with injected anomalies such as frequency shifts. To demonstrate real-world applicability, publicly available Kepler light curve data was incorporated in later stages of development [9].

3.2 Preprocessing

Preprocessing is identified as a critical step in the pipeline. Time-series signals are normalized to stabilize input distributions while preserving variations relevant to anomaly detection.

We considered multiple spike encoding strategies, including temporal and rate-based encoding. We selected rate encoding for initial implementation due to its simplicity and compatibility with existing SNN frameworks. This encoding converts continuous signals into spike trains by representing each value through firing frequency. While it served as an efficient process, the rate encoding was too simplistic and failed to capture varied inputs (such as different types of anomalies) during training.

To address these issues, we decided to work with delta encoding instead, which focused on encoding changes in the input signal instead of absolute values. Delta encoding provided a responsive representation, improving the model's temporal precision and dynamic pattern detection.

3.3 Training and Evaluation

To test the robustness of the proposed model, multiple synthetic anomaly types were introduced into the time-series data. These anomalies were designed to reflect different ways real astronomical signals could possibly deviate from normal behavior. These include:

- **Amplitude Burst:** Sudden spikes in signal magnitude that represent abrupt increases in intensity.
- **Frequency Shift:** Changes in oscillation rate, causing a signal to become faster or slower than normal.
- **Phase Shift:** Temporal displacement of a signal pattern, causing shifted waveform alignment.
- **Noise Burst:** Localized regions of high random noise that disrupt underlying signal structuring.
- **Dropout:** Missing or flattened segments of the signal, showing data loss or sensor failure.

The SNN model was trained using only normal data to learn baseline temporal dynamics, following a semi-supervised anomaly detection approach. Anomalies are identified as deviations from expected spike patterns.

For each test signal, the model computes a reconstruction score based on how closely the output matches the learned normal pattern. This score is then compared against a predefined threshold. If the reconstruction score is below the threshold, the signal is classified as an anomaly; otherwise, it is classified as normal.

Lastly, performance is evaluated using detection rate, false alarm rate, and anomaly scoring consistency.

4 Results

4.1 Single Anomaly Detection and Threshold Analysis

Encoding Threshold	Accuracy	Precision	Recall	F1
0.03	0.6250	0.6667	0.5000	0.5714
0.05	0.6750	0.6842	0.6500	0.6667
0.08	0.5500	1.0000	0.1000	0.1818

Table 1: Performance metrics for different encoding thresholds.

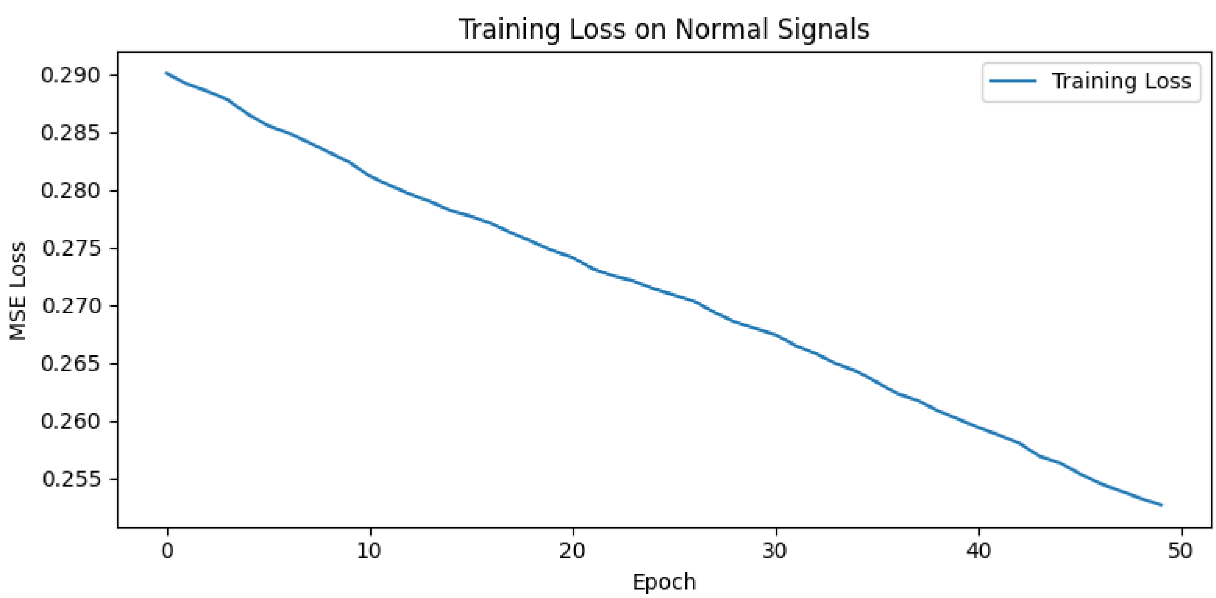

Figure 1: MSE Loss across 50 epochs (0.05 threshold).

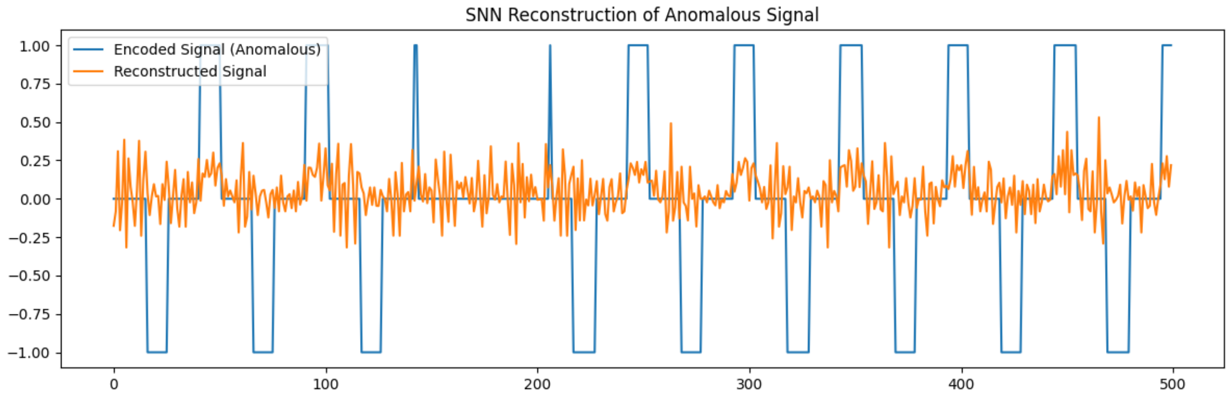

Figure 2: Delta-encoding of normal signals (0.05 threshold).

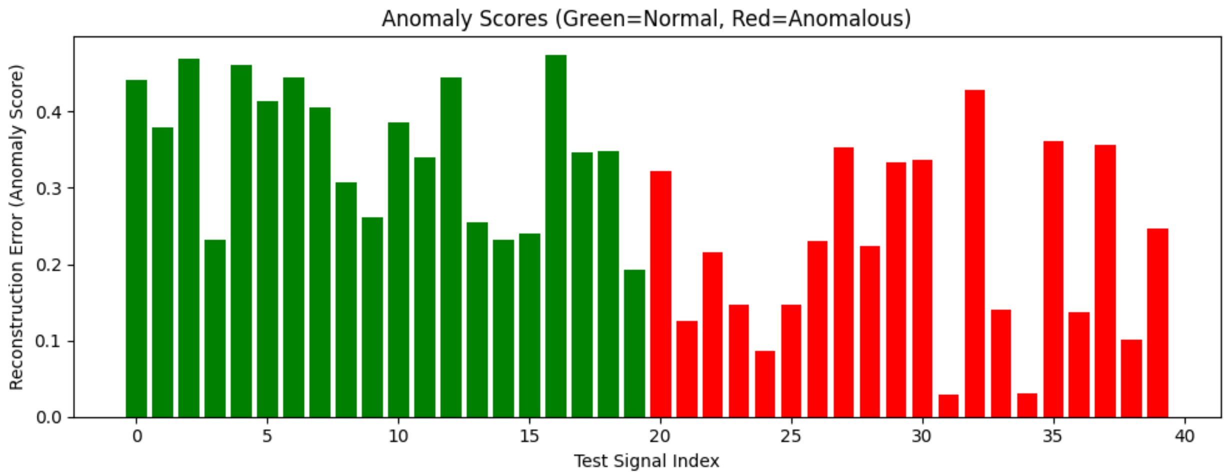

Figure 3: Anomaly scores (0.05 threshold).

4.2 Multi-Anomaly Detection

Experiment	Threshold	TP	TN	FP	FN	Accuracy	Precision	Recall	F1
Amplitude Burst	0.2128	20	18	2	0	0.9500	0.9091	1.0000	0.9524
Frequency Shift	0.2254	1	17	3	19	0.4500	0.2500	0.0500	0.0833
Phase Shift	0.2327	3	16	4	17	0.4750	0.4286	0.1500	0.2222
Noise Burst	0.2389	19	16	4	1	0.8750	0.8261	0.9500	0.8837
Dropout	0.2199	5	16	4	15	0.5250	0.5556	0.2500	0.3448

Table 2: Performance metrics across different anomaly types and threshold settings.

4.3 Kepler Light Curve Case Study

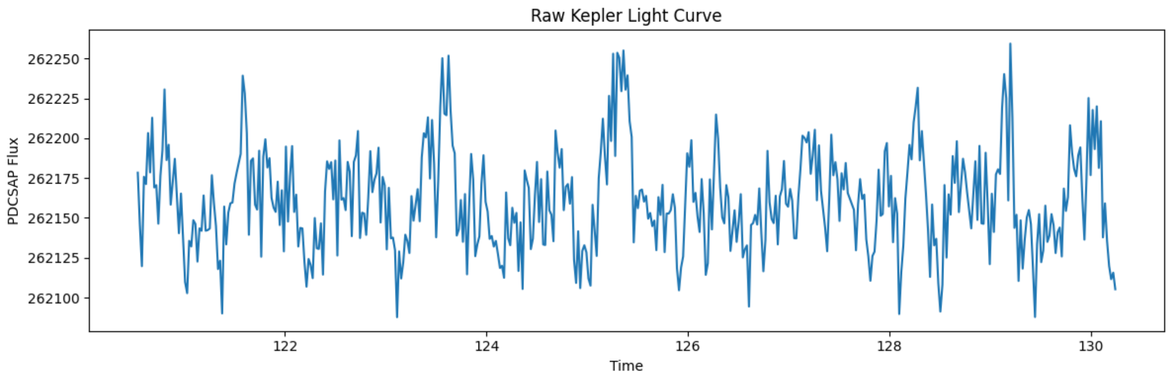

Figure 4: Raw Kepler light curve.

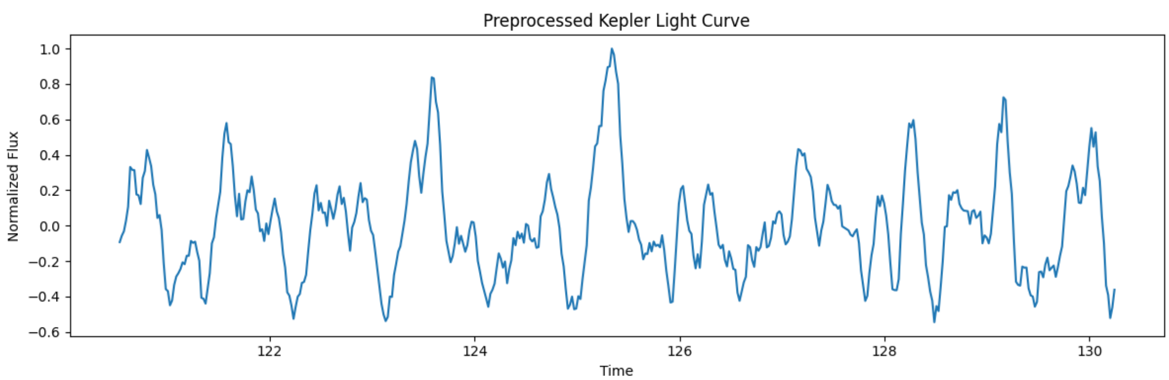

Figure 5: Preprocessed Kepler light curve after normalization, smoothing, and scaling.

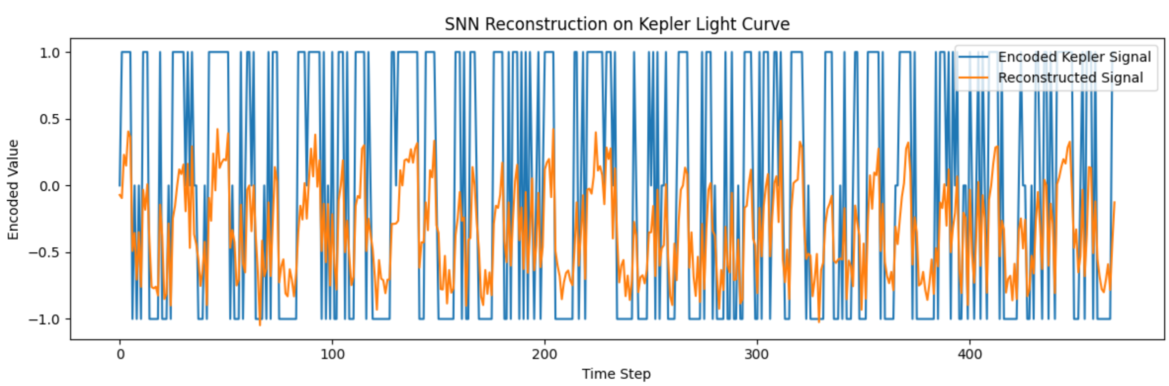

Figure 6: SNN reconstruction on the encoded Kepler light curve using adaptive encoding.

5 Evaluation

To evaluate the pipeline on real astronomical data, we applied the trained SNN model to a publicly available Kepler light curve from the NASA Kepler Mission Archive. Unlike the synthetic experiments, the Kepler data did not contain ground-truth anomaly labels, so this experiment was used as a qualitative transfer test rather than a fully supervised evaluation.

The raw Kepler light curve is shown in Figure 4. The first attempt to implement the model using this raw signal resulted in a poor reconstruction performance. This occurred because the raw Kepler flux values were much larger than the normalized synthetic signals used during training. As a result, the spike encoder generated dense and noisy spike patterns, and the reconstructed signal had little identifiable structure. The initial reconstruction error was approximately 0.93 MSE.

In order to solve this problem, the Kepler light curve was normalized, filtered, and rescaled before encoding. The effect of this preprocessing step is shown in Figure 5, where the signal is transformed into a normalized range while still preserving its major temporal variations. This preprocessing step reduced the effect of raw magnitude differences and made the real data more consistent with the synthetic training distribution. In addition, a fixed encoding threshold was replaced with an adaptive threshold based on the standard deviation of the preprocessed Kepler signal. The final Kepler adaptive encoding threshold was 0.014557.

After preprocessing and adaptive encoding, the reconstruction MSE decreased to 0.540704. The final reconstruction result is shown in Figure 6. Although the reconstruction remained imperfect, the model was able to process real light curve data and produce a structured output. This result demonstrates that the proposed pipeline can transfer to real astronomical time-series data, while also highlighting the limitations of simple spike encoding when applied to more complex real-world signals.

6 Discussion

The model was trained for 50 epochs, with the loss steadily decreasing, shown in Figure 1. This shows that the model gradually improved its ability to reconstruct normal time-series signals. Initially, synthetic testing with a single anomaly type produced nearly-perfect separation. However, this behavior was not exactly accurate, as the original anomaly samples had a simple pattern. While this demonstrated a strong baseline temporal structure, it did not fully consider other anomaly detection conditions.

Thus, in the next phase of implementation, we introduced multiple anomaly types. This caused the model's performance to be more realistic, as there was some overlap between the normal and anomalous reconstruction scores. As seen in Table 2, the performance became much more variable across different anomaly categories. This introduced another challenge behind the generalization of anomaly detection across different temporal behaviors.

From these two stages, it became clear that the model's performance depends heavily on the encoding strategy and threshold selection. In this project, the threshold value determines when a change in the input signal is significant enough to generate a spike. As we tested delta-based encoding with different threshold values on simple anomalies, we found that out of 0.03, 0.05, and 0.08, a threshold of 0.05 provided the best overall score, shown in Table 1. Lower thresholds tended to introduce more noise, while high thresholds would create higher sensitivity.

Additionally, the results show that model performance varies depending on the type of anomaly being evaluated. The model shows to performs better on magnitude-based anomalies, such as amplitude bursts, noise bursts, and dropouts, while performance decreases for timing-based anomalies, such as frequency and phase shifts. This difference suggests that the current SNN configuration is more sensitive to changes in signal intensity than to temporal misalignment.

The Kepler case study further proved the importance of the preprocessing and encoding steps. When the raw Kepler light curve was first passed into the model, the reconstruction quality was poor, since the scale of Kepler's real flux values were very different from the synthetic training signals. After normalization, smoothing, and thresholding adjustments, the reconstruction MSE improve, although not perfectly. This result suggests that delta encoding still loses important amplitude and timing details within each signal.

Overall, the results demonstrate that delta encoding combined with tuned thresholds provide a stronger framework for anomaly detection in SNNs. However, results across multiple anomaly types and real-world signals suggest that further improvements in encoding design and model architecture are needed for stronger generalization abilities.

7 Future Work

Future extensions of this project could involve exploring other advanced spike encoding techniques. While we experimented with spike encoding and delta encoding, other schemes, such as learnable or hybrid encoding strategies that combine both rate and delta-based representations, could be experimented with. This could allow the model to better track global signal intensity and smaller temporal changes, causing an improvement in anomaly detection.

Another important extension would be to explore alternative SNN architectures. This includes deeper recurrent spiking models and attention-based spiking networks, which could assist with detecting longer-range temporal dependencies in astronomical signals instead of more "isolated" anomalies.

Finally, future work could include evaluation on larger, more diverse astronomical datasets beyond Kepler. For example, datasets that contain anomalies from multiple sources would assist with determining the model's generalization abilities.

8 References

[1] “Time-Domain Astronomy,” Harvard-Smithsonian Center for Astrophysics. https://www.cfa.harvard.edu/research/topic/time-domain-astronomy.

[2] V. Etsebeth, M. Lochner, M. Walmsley, and M. Grespan, “Astronomaly at scale: Searching for anomalies amongst 4 million galaxies,” Monthly Notices of the Royal Astronomical Society, vol. 529, no. 1, pp. 732–747, Mar. 2024. https://arxiv.org/pdf/2309.08660

[3] K. Yamazaki et al., ”Spiking neural networks and their applications: A review,” Brain Sciences, vol. 12, no. 7, p. 863, Jun. 2022. doi: https://doi.org/10.3390/brainsci12070863.

[4] N. J. Pritchard, A. Wicenec, R. Dodson, and M. Bennamoun, “Polarisation-inclusive spiking neural networks for real-time RFI detection in modern radio telescopes,” arXiv preprint arXiv:2504.11720v2, 2025. arXiv:2504.11720

[5] H. Sabbella, A. Mukherjee, T. Kandappu, S. Dey, A. Pal, A. Misra, and D. Ma, “The promise of spiking neural networks for ubiquitous computing: A survey and new perspectives,” arXiv preprint arXiv:2506.01737, 2025. https://arxiv.org/pdf/2506.01737

[6] D. Muthukrishna, G. Narayan, K. S. Mandel, R. Biswas, and R. Hložek, ”RAPID: Early Classification of Explosive Transients Using Deep Learning,” Publications of the Astronomical Society of the Pacific, vol. 131, no. 1005, p. 118002, Sep. 2019. doi: 10.1088/1538-3873/ab1609. https://doi.org/10.1088/1538-3873/ab1609

[7] N. J. Pritchard, A. Wicenec, M. Bennamoun, and R. Dodson, ”Spiking neural networks for radio frequency interference detection in radio astronomy,” Communications Physics, vol. 8, p. 517, 2025. doi: 10.1038/s42005-025-02420-7. https://doi.org/10.1038/s42005-025-02420-7

[8] Bäßler, D., Kortus, T., and Gühring, G, “Unsupervised anomaly detection in multivariate time series with online evolving spiking neural networks,” Machine Learning, vol. 111, pp 1377-1408, 2022, doi:10.1007/s10994-022-06129-4

[9] NASA Kepler Mission Archive, Mikulski Archive for Space Telescopes (MAST), Space Telescope Science Institute. https://archive.stsci.edu/

9 Appendix

Link to GitHub repository: https://github.com/akalyasri/Astronomical-Anomaly-Detection-with-Spiking-Neural-Networks